

Chronic Canaliculitis (MONDO:0004924): A Comprehensive Disease Characteristics Report

Summary

Chronic canaliculitis is an uncommon, indolent, acquired infection of the lacrimal canaliculus—the small epithelial-lined channel that drains tears from the eyelid margin toward the lacrimal sac. It is caused predominantly by the filamentous, Gram-positive anaerobe *Actinomyces* (frequently in polymicrobial combination), which aggregates within the canalicular lumen to form calcified concretions ("canaliculiths") and biofilm. The disease manifests as chronic epiphora (tearing), mucopurulent discharge, a red and "pouting" lacrimal punctum, and medial eyelid inflammation. It is overwhelmingly a disease of **middle-aged and elderly women**, is almost always **unilateral**, and most often affects the **lower canaliculus**. Because its signs mimic more common conditions, it is **frequently and repeatedly misdiagnosed** as bacterial/viral conjunctivitis, chronic dacryocystitis, or chalazion, producing diagnostic delays that commonly reach 10–30 months.

Mechanistically, the disease is not genetic. It arises from a self-perpetuating cycle of **tear stasis → organism colonization → intracanalicular concretion/biofilm formation → chronic mucosal inflammation → further stasis and occlusion**. Age-related involutional changes of the drainage system (punctal atrophy, canalicular fibrosis), dry-eye disease, an odontogenic *Actinomyces* reservoir, and iatrogenic punctal/canalicular plugs used to treat dry eye are the principal predisposing/risk factors. There are no established causal genes, no inheritance, no animal models, and no omics/biomarker datasets specific to this disease—reflecting its status as a localized, acquired, infectious/inflammatory condition rather than a Mendelian or systemic disorder.

Prognosis is **excellent once the concretions are physically removed**. Definitive treatment is canaliculotomy with curettage of concretions, with reported cure rates of ~92–98%, further improved by adjunctive silicone tube intubation and punctum-sparing techniques. Conservative, incision-sparing management (punctal dilation, manual expression, microcurettage, and antibiotic canalicular irrigation) is an effective first-line alternative in

selected cases (~83% success). There is no disease-specific mortality; morbidity is limited to local discomfort, recurrent infection, and—if untreated—canalicular dilatation and stenosis. The most important diagnostic principle is clinical vigilance: recognizing the "pouting punctum" and canalicular discharge, supported by high-frequency ultrasound biomicroscopy showing intraluminal concretions, and confirming with expression/curettage plus microbiology/histopathology (which also excludes the rare masquerade of canalicular carcinoma).

1. Disease Information

Overview. Chronic canaliculitis is a chronic, low-grade infection/inflammation of the lacrimal canaliculus. The canaliculi are the initial, epithelial-lined segments of the lacrimal drainage system beginning at the punctum on each eyelid margin. Primary (or "primary chronic") canaliculitis denotes intrinsic infection of the canaliculus, typically associated with concretions; secondary canaliculitis denotes infection provoked by a retained foreign body, most notably a punctal/canalicular plug. The condition is characterized by its indolent course, its tendency to form intraluminal concretions, and its notoriety for misdiagnosis.

"Primary chronic canaliculitis is an uncommon disease, which is often misdiagnosed and insufficiently treated." — [PMID: 15764110](#)

"Primary lacrimal canaliculitis (PLC) is a unique disorder which often gets misdiagnosed by the general as well as speciality-trained ophthalmologists." — [PMID: 29564416](#)

Key identifiers.

Resource	Identifier
MONDO	MONDO:0004924 (canaliculitis)
ICD-10	H04.3 (acute and unspecified inflammation of lacrimal passages); H04.4 (chronic inflammation of lacrimal passages)
ICD-11	9A27.1 (inflammation of lacrimal passages / canaliculitis)
MeSH	Lacrimal Apparatus Diseases (canaliculitis indexed under lacrimal apparatus disease terms)
OMIM	Not applicable (no Mendelian entry — acquired infectious disease)
Orphanet	Not a listed rare Mendelian disorder
SNOMED CT	Canaliculitis (disorder)

Synonyms / alternative names. Lacrimal canaliculitis; primary canaliculitis; primary chronic canaliculitis (PCC); primary lacrimal canaliculitis (PLC); suppurative canaliculitis; chronic suppurative canaliculitis; Actinomycotic canaliculitis (when *Actinomyces* is confirmed). A clinically overlapping but distinct entity is "idiopathic canicular inflammatory disease" ([PMID: 29373404](#)).

Source of information. The knowledge base for this disease is derived overwhelmingly from **aggregated disease-level resources**—retrospective single-center and multicenter case series and case reports—rather than large EHR cohorts or population registries. This reflects its rarity. The largest series comprise tens to a few hundred patients (e.g., 74, 201, 338 cases).

2. Etiology

Primary causal factors. Chronic canaliculitis is an **infectious/inflammatory acquired disease**, not a genetic one. The dominant causal organism is *Actinomyces israelii* and related *Actinomyces* species—filamentous, Gram-positive, cast-forming anaerobes that aggregate to form canicular concretions. The infection is frequently **polymicrobial**.

"Histological examination identified Actinomyces species in 11 of 15 concretions (73%; 95% confidence interval, Wilson method: 48 - 89%), supporting their leading role as causative organisms of canaliculitis." — [PMID: 41702562](#)

Risk factors — environmental / acquired. - **Age** (older adults; mean ages 48–66 y across series) and **female sex** are the strongest demographic risk factors. - **Dry-eye disease** is a major predisposing factor, both directly (reduced tear clearance/flushing) and indirectly (leading to plug insertion). - **Punctal/canalicular plugs** inserted for dry eye are a key **iatrogenic** risk factor for secondary canaliculitis.

"Dry eye was identified in the vast majority of patients with Actinomycotic canaliculitis. Most cases are odontogenic in origin and the infection occurs in immunocompetent individuals." — [PMID: 36927124](#)

- **Odontogenic reservoir:** *Actinomyces* is part of normal oral/periodontal flora; an odontogenic source is common.
- **Involutional anatomical changes:** punctal atrophy, canalicular fibrosis, and nasolacrimal duct stenosis promote tear stagnation ([PMID: 42366665](#)).

Genetic risk factors. None established. No susceptibility loci, modifier genes, or GWAS associations exist for this localized acquired infection.

Protective factors. No genetic protective variants are known. Environmentally, adequate tear drainage, avoidance/early removal of punctal plugs, and good oral/periodontal hygiene are logical (though not formally studied) protective measures. Hosts are typically **immunocompetent**, so immune status is not a strong determinant for primary disease (in contrast to fungal cases).

Gene–environment interactions. Not applicable — no genetic contribution has been identified.

3. Phenotypes

Chronic canaliculitis presents with a characteristic cluster of **symptoms and clinical signs** (not laboratory or behavioral abnormalities). The proposed diagnostic "**clinical tetrad**" captures the core phenotype:

"We propose a 'clinical tetrad' of 1. medial eyelid edema, 2. pouting and hyperemia of lacrimal punctum, 3. yellowish canalicular hue and, 4. canalicular distention, and expressible discharge, for the easier clinical diagnosis of LC." — PMID: 36644466

Phenotype	Type	Frequency	HPO suggestion
Epiphora (excessive tearing)	Symptom	~85–90% (63/74; 90.5% in canaliculoplasty series)	HP:0009926 (Epiphora)
Mucopurulent discharge from punctum	Sign	~85% (85.7%)	Ocular discharge
Pouting/hyperemic punctum	Sign	Highly characteristic (tetrad component)	—
Medial eyelid/canthal edema	Sign	Common (tetrad component)	HP:0000534 (Abnormal eyelid morphology, broad)
Canalicular concretions (canaliculiths)	Physical manifestation	Large majority (15/17; 37/37)	—
Canalicular distention/dilatation	Sign	Subset (severe/long-standing)	—
Redness/irritation (mimics conjunctivitis)	Symptom/sign	Common	HP:0000509 (Conjunctivitis)

"The most common presenting symptom was epiphora, noted in 63 (85%) patients" — PMID: 22836798

Phenotype characteristics. - **Age of onset:** adult-onset, typically middle-aged to geriatric (mean 48–66 y). Pediatric onset is *extremely rare* (PMID: 31055896). - **Severity:** mild-to-moderate and localized; rarely sight-threatening. - **Progression:** chronic, indolent, episodic/recurrent; can progress to canalicular dilatation and stenosis if untreated. - **Frequency among affected individuals:** epiphora and discharge are near-universal; concretions are present in the large majority.

Quality-of-life impact. Chronic tearing, recurrent discharge, and repeated ineffective treatment (often over months to years) impose meaningful ocular-surface discomfort and repeated clinic visits, but the disease does not threaten vision or life. No disease-specific EQ-5D/SF-36 data exist.

4. Genetic / Molecular Information

Not applicable. Chronic canaliculitis is an acquired infectious disease with **no causal genes, no pathogenic germline or somatic variants, no modifier genes, no epigenetic signatures, and no chromosomal abnormalities**. There are no OMIM, ClinVar, HGMD, or gnomAD entries relevant to disease causation. Reports of neoplasia (plasmacytoma, carcinoma) associated with or masquerading as canaliculitis are coincidental mass lesions, not a genetic basis for canaliculitis itself (PMID: 21743365; PMID: 16534063).

5. Environmental Information

Environmental factors. The relevant "environmental" exposures are microbiological and iatrogenic rather than chemical/toxic: - **Iatrogenic foreign bodies:** punctal and intracanalicular plugs (e.g., SmartPLUG) used for dry-eye management. Plug-related canaliculitis represented **18.3%** of all canaliculitis cases in one series; all affected patients were female with prior plug insertion.

"Cultures of discharge, concretions, and/or infected plugs mostly revealed Pseudomonas aeruginosa (42%)." — PMID: 34139956

Lifestyle factors. Poor oral/periodontal hygiene plausibly contributes via the odontogenic *Actinomyces* reservoir, though this is inferential.

Infectious agents (the core etiology).

Category	Key organisms	Evidence
Bacteria (dominant)	<i>Actinomyces israelii</i> /spp. (leading), <i>Staphylococcus</i> spp. (39% most common culture isolate), <i>Streptococcus</i> spp., <i>Nocardia</i> spp., polymicrobial anaerobes	PMID: 41702562 ; PMID: 22836798 ; PMID: 34287258
Bacteria (plug-related)	<i>Pseudomonas aeruginosa</i> (42%)	PMID: 34139956
Rare/novel bacteria	<i>Arcanobacterium haemolyticum</i> , <i>Tsukamurella</i> spp., <i>Ottowia massiliensis</i> , <i>Fusobacterium periodonticum</i>	PMID: 15764110 ; PMID: 31246677 ; PMID: 40234845 ; PMID: 41429755
Fungi	<i>Candida</i> (27.1%), <i>Aspergillus</i> (23.7%), <i>Scedosporium</i> (novel)	PMID: 42782278 ; PMID: 40788664
Viruses	HSV (83.1% of viral cases), VZV	PMID: 41528827 ; PMID: 29426966

"The most commonly reported species were *Candida* (27.1%) and *Aspergillus* (23.7%)."
[PMID: 42782278](#)

"The majority of viral lacrimal drainage infections were secondary to herpes simplex virus (HSV) (83.1%, 378/455)" — [PMID: 41528827](#)

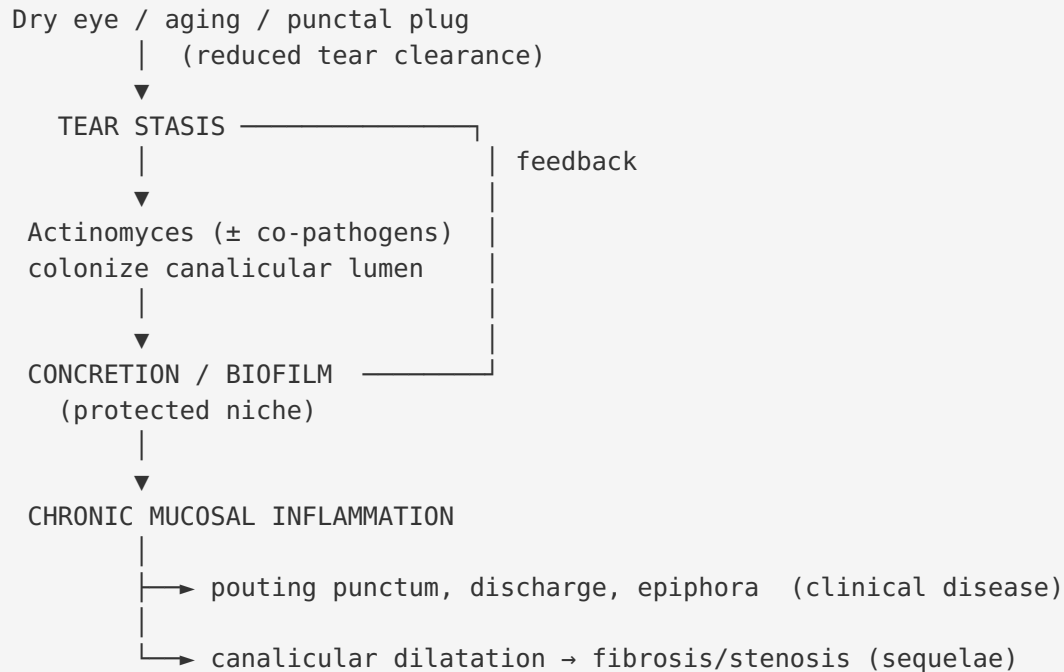
Suggested NCBITaxon anchors: *Actinomyces israelii* (NCBITaxon:1659); *Pseudomonas aeruginosa* (NCBITaxon:287); *Candida albicans* (NCBITaxon:5476); *Human alphaherpesvirus 1* (NCBITaxon:10298).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Predisposing anatomical/physiological change** (age-related punctal atrophy, canalicular fibrosis, dry eye, or an inserted punctal plug) **leads to** reduced tear flow and **tear stasis** within the canaliculus.
2. Tear stasis **results in** colonization of the canalicular lumen by resident/odontogenic organisms, chiefly *Actinomyces* (often with co-pathogens).
3. Colonizing filamentous organisms **aggregate and mineralize**, which **leads to** formation of intracanalicular **concretions (canaliculiths) and biofilm**. (*Demonstrated: concretions present in the large majority of cases and shown histologically to contain aggregated Actinomyces/anaerobes.*)
4. The concretion/biofilm **results in** a protected microbial niche that resists topical antibiotics and **sustains chronic mucosal inflammation** of the canalicular epithelium.
5. Chronic inflammation **leads to** the clinical phenotype: pouting hyperemic punctum, mucopurulent discharge, medial eyelid edema, and epiphora.
6. Persistent inflammation **further results in** post-inflammatory sequelae—canalicular dilatation and, ultimately, **fibrosis/stenosis**—which worsen stasis and **feed back** to step 2, closing a self-perpetuating loop. (*Branch: if a foreign body/plug is present, it substitutes for steps 1–3 as the nidus.*)

"Lacrimal infection is a vicious circle, in which infection leads to inflammation and post-inflammatory sequelae, themselves a source of occlusion and stagnation, which in turn encourages infection." — [PMID: 39488146](https://pubmed.ncbi.nlm.nih.gov/39488146/)



Cellular processes & immune involvement. The core process is **chronic infection-driven inflammation** (GO:0006954, inflammatory response; GO:0006935, chemotaxis of neutrophils) of the **canalicular epithelium**, with mixed suppurative/granulomatous features histologically and no autoimmune basis (collagen-vascular/autoimmune screens negative in the related idiopathic entity). Hosts are typically immunocompetent for bacterial disease; immunosuppression is relevant chiefly for fungal cases.

Biofilm formation (GO:0042710) is the pivotal cellular-community process explaining chronicity and antibiotic tolerance. **Concretion mineralization** represents biomineral aggregation of filamentous organisms.

Molecular pathways / metabolic changes / protein dysfunction. No specific host signaling cascade (Wnt, MAPK, mTOR, PI3K-AKT), enzyme deficiency, metabolic derangement, or protein misfolding is implicated—consistent with an acquired, localized infection rather than a molecular-genetic disease.

Molecular profiling. No transcriptomic, proteomic, metabolomic, or lipidomic disease signatures exist. The only "omics" applied is **metagenomic shotgun sequencing of concretions** for pathogen identification, which detected bacteria (predominantly anaerobes) in all sampled concretions ([PMID: 34287258](https://pubmed.ncbi.nlm.nih.gov/34287258/)).

Suggested ontology terms. Biological processes: GO:0006954 (inflammatory response), GO:0042710 (biofilm formation), GO:0009617 (response to bacterium). Cell types: CL:0000066 (epithelial cell) of the canalicular lining; CL:0000775 (neutrophil). Anatomy: UBERON:0002392 (lacrimal canaliculus).

7. Anatomical Structures Affected

Organ / system level. Primary organ: the **lacrimal drainage apparatus**, specifically the **lacrimal canaliculus** (UBERON:0002392). Body system: the ocular adnexa / lacrimal (nasolacrimal) drainage system of the visual system. Secondary involvement: the lacrimal sac and nasolacrimal duct can be secondarily affected in advanced or overlapping disease (dacryocystitis differential).

Localization and laterality. The **lower (inferior) canaliculus** is most often affected, and disease is overwhelmingly **unilateral**.

"Lower canaliculus was involved in 48 (65%) patients, upper canaliculus in 17 (23%) patients, and both canaliculi in 9 (12%) patients." — PMID: 22836798

Tissue / cell level. The affected tissue is the **epithelial lining of the canaliculus** (stratified/columnar epithelium) and adjacent subepithelial stroma, with inflammatory cell infiltration. Concretions occupy the lumen.

Subcellular level. No specific organelle/compartment pathology; suggested GO cellular component anchors are extracellular (biofilm matrix; GO:0005576, extracellular region) rather than intracellular.

Suggested ontology terms. UBERON:0002392 (lacrimal canaliculus); UBERON:0001817 (lacrimal apparatus); UBERON:0000970 (eye, broad); CL:0000066 (epithelial cell).

8. Temporal Development

Onset. Adult-onset, insidious, and **chronic**. Mean age at presentation ranges 48–66 years; pediatric onset is extremely rare.

"It is extremely rare in children and infants." — PMID: 31055896

Progression and course. The disease is chronic, indolent, and episodic/recurrent. Diagnostic delay is a hallmark, with mean time-to-diagnosis of ~10 months in one series and **30.6 ± 39.5 months** in a canaliculoplasty series of long-standing disease. Untreated disease can progress from mucosal inflammation to **canalicular dilatation** and ultimately **stenosis/obstruction**.

Disease stages (as described for the related idiopathic canalicular inflammatory disease, staged 1–5): edema → progressive centripetal vascularization → pouting of vascularized mucosa → membrane formation → progressive scarring (PMID: 29373404). Classic infectious chronic canaliculitis does not have a formal universally adopted staging system.

Remission / critical periods. Remission is **treatment-induced** (concretion removal), not spontaneous—residual concretions perpetuate disease. The therapeutic "critical window" is the point of definitive concretion removal before irreversible fibrosis/stenosis develops.

Duration. Chronic and persistent until definitively treated; not self-limited.

9. Inheritance and Population

Epidemiology. Chronic canaliculitis is a **rare** lacrimal disorder (historically cited as ~2% of lacrimal disease). Precise population incidence/prevalence figures are not established due to under-recognition and misdiagnosis. Among **primary** canaliculitis, *Actinomyces* accounts for roughly 11% of cases (22/201).

"Of the 201 patients diagnosed with primary canaliculitis, 22 (10.9%) were caused by [Actinomyces]" — PMID: 36927124

Inheritance. Not applicable — acquired infectious disease with no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency.

Population demographics. - **Sex ratio:** female predominance is consistent and often marked—54% female in the largest primary series, rising to 77–83% in others; plug-related cases are essentially all female. - **Age distribution:** middle-aged to elderly.

"Of the 74 patients, 40 (54%) were women. Mean age at presentation was 48 years." — PMID: 22836798

- **Geographic distribution:** worldwide; case series originate from Asia, Europe, North America, and elsewhere, with no strong endemic clustering. No variant geography (no variants).

10. Diagnostics

Diagnosis is fundamentally clinical, confirmed by expression/curettage of concretions with microbiology and histopathology.

Clinical signs / criteria. The "**clinical tetrad**" (medial eyelid edema; pouting/hyperemic punctum; yellowish canalicular hue; canalicular distention with expressible discharge) enables bedside diagnosis (PMID: 36644466). Expression of concretions/discharge from the punctum is essentially pathognomonic.

Imaging. High-frequency (80-MHz) **ultrasound biomicroscopy (UBM)**, optionally combined with color Doppler flow imaging, demonstrates luminal ectasia with a high-echo intraluminal mass (calculi) and uneven mucosal thickening.

"Lacrimal canaliculitis (vertical section) showed obvious ectasia of the lacrimal canalicular lumen, with a high echo mass shadow, which might have been calculi" — PMID: 31823059

Adjunctive tools include **dacryoendoscopy** and Fourier-domain OCT (used in the idiopathic inflammatory variant).

Microbiology / pathology. Culture (aerobic + anaerobic), Gram stain, and histopathology of curetted concretions establish the organism; *Actinomyces* appears as sulfur-granule-like filamentous colonies. **Metagenomic/next-generation sequencing** can identify fastidious or unculturable pathogens directly from concretions.

"Sequencing analysis detected bacteria in all samples" — PMID: 34287258

Genetic / omics testing. Not applicable for diagnosis (no genetic basis; no validated biomarkers). Molecular testing is limited to pathogen identification (16S rDNA sequencing, PCR for viral cases).

Differential diagnosis (critical). Chronic dacryocystitis, chalazion/hordeolum, and conjunctivitis are the leading mimics; rare masquerades include canalicular carcinoma and plasmacytoma. Histopathology of excised tissue is important in atypical/refractory cases.

"A history of chronic redness, watering, discharge, and medial canthal region edema lead to the misdiagnosis of chronic dacryocystitis in 3 (60%) and medial marginal chalazion in 2 (40%) cases." — PMID: 29564416

"Carcinoma of the lacrimal canaliculus masquerading as canaliculitis" — PMID: 16534063

Screening. No population/newborn/carrier screening applies (acquired, non-genetic, rare).

11. Outcome / Prognosis

Prognosis is excellent once concretions are removed. There is **no disease-specific mortality** and no threat to life; morbidity is local.

Outcome metric	Value	Source
Cure by canaliculotomy (large dacryolithiasis series)	297/302 = 98.34%	PMID: 40067167
Recurrence-free at 3.7 y (canaliculotomy + silicone tube)	88%	PMID: 36431305
Complete remission (canaliculoplasty for dilated canaliculus)	33/42 = 78.6%	PMID: 32563241
Success, incision-sparing management	10/12 = 83.3%	PMID: 28576205

"Of 302 cases (89.35%) with canaliculitis, 297 (98.34%) were cured with canaliculotomy" — PMID: 40067167

Complications. Recurrent/persistent infection if concretions are incompletely removed; canalicular dilatation; **canalicular stenosis/obstruction** (a recognized post-surgical and post-inflammatory complication—3 patients developed stenosis in the canaliculoplasty series). Punctum-damaging surgery risks punctal deformity, motivating punctum-sparing techniques.

Prognostic factors. Completeness of concretion removal is the dominant prognostic determinant. Long-standing disease with canalicular dilatation is a more severe phenotype. No molecular prognostic biomarkers exist.

Quality of life / disability. Limited to chronic ocular-surface discomfort and epiphora; no formal QoL instrument data. Full functional recovery is expected after definitive treatment.

12. Treatment

Treatment aims to **eradicate the microbial nidus by physically removing concretions**, since topical antibiotics alone frequently fail to penetrate the biofilm/concretion.

Tiered treatment strategy

1. Conservative / incision-sparing (first-line in selected cases): punctal dilation, manual expression/massage of concretions, microcurettage, and **canalicular irrigation with susceptible antibiotics** (e.g., fortified cefazolin, ciprofloxacin).

"Ten (83.3%) eyes were successfully treated with incision-sparing modalities, and 2 (16.7%) eyes were treated surgically. No recurrences were observed" — PMID: 28576205

"The conservative method combining canalicular expression and irrigation with topical susceptible antibiotics is recommendable as initial therapy." — PMID: 36927124

Intracanalicular antibiotics can obviate surgery in some suppurative cases (PMID: 18580001).

2. Definitive surgery — canaliculotomy with curettage: the mainstay for refractory/recurrent disease, with cure rates ~92–98%.

3. Adjuncts improving outcomes: - **Silicone tube intubation** improves anatomical/functional success and long-term recurrence-free rates.

"a higher success rate can be achieved when silicone tube intubation is performed during the procedure" — PMID: 34275398

"After a follow-up time of 3.7 ± 1.5 years, 88% of cases showed no recurrence of inflammation." — PMID: 36431305

- **Punctum-sparing / mini-invasive canaliculotomy** preserves punctal function with high success (anatomic 96–98%, functional 91–94%) and no recurrences (PMID: 36473974; PMID: 37300269; PMID: 42530488).
- **Canaliculoplasty with stenting** for the severe dilated-canalculus phenotype (PMID: 32563241).

4. Pathogen-directed antimicrobials: systemic/topical penicillin for *Actinomyces*; topical/systemic **azoles/voriconazole** for fungal (e.g., *Scedosporium*, *Candida*, *Aspergillus*) cases; topical **acyclovir** for HSV/VZV canaliculitis; plug removal for secondary/plug-related disease.

5. Secondary (plug-related) canaliculitis: remove the offending plug (office irrigation, retrograde massage, or canaliculotomy); DCR may be needed if obstruction persists (PMID: 16920195).

Suggested NCIT anchors: NCIT:C15329 (Antibiotic Therapy); surgical canaliculotomy (ophthalmologic surgical procedure); Voriconazole; Acyclovir.

Pharmacogenomics / advanced therapeutics (gene, cell, RNA, targeted, immuno-therapy): Not applicable.

13. Prevention

- **Primary prevention:** manage dry eye without unnecessary permanent plugs; judicious plug selection and monitoring; good oral/periodontal hygiene to limit the *Actinomyces* reservoir. These are logical but not formally trialed.
- **Secondary prevention:** early recognition of the clinical tetrad to avoid diagnostic delay; prompt removal of retained plugs at first sign of inflammation.
- **Tertiary prevention:** complete concretion removal and adjunctive intubation to prevent recurrence and canalicular stenosis.

- **Immunization, genetic screening/counseling, public-health/environmental interventions, prophylactic medication:** Not applicable.
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14. Other Species / Natural Disease

Not applicable / not reported. Chronic canaliculitis as described here is a human disease of the lacrimal drainage apparatus. There is no established naturally occurring counterpart catalogued in veterinary resources (OMIA) for this specific entity, no breed predisposition, no orthologous causal gene (there is no causal gene), and no zoonotic transmission. *Actinomyces* and related organisms exist across species, but canalicular concretion disease is not a recognized comparative-pathology entity in the reviewed literature.

Suggested taxonomy anchor for the host: *Homo sapiens* (NCBITaxon:9606).

15. Model Organisms

None exist. There are no mouse, rat, zebrafish, invertebrate, cellular, organoid, or iPSC models of chronic canaliculitis in the reviewed literature. This absence reflects the disease's nature as a localized, acquired human infection driven by concretion/biofilm formation—features not readily recapitulated in standard genetic model systems. Consequently there is no phenotype-recapitulation or model-limitation data to report, and no model databases (MGI, RGD, ZFIN, etc.) hold relevant entries. This is a clear knowledge gap: an in vitro **biofilm/concretion model** using canalicular epithelial cells co-cultured with *Actinomyces* would be a logical future development.

Key Findings (Consolidated with Evidence)

Finding 1 — A rare, misdiagnosed infection with female predominance and older-adult onset

Across the largest primary-canalculitis series (74 patients), 54% were women, mean age 48 years, epiphora was the leading symptom (85%), the lower canaliculus was involved in 65%, and the mean diagnostic delay was ~10 months; other series report even higher female predominance (77–83%) and older mean ages (57–63 y). Misdiagnosis is the rule rather than the exception. (Evidence: [PMID: 22836798](#), [PMID: 29564416](#))

Finding 2 — *Actinomyces* is the leading cause; disease is frequently polymicrobial and concretion-driven

Histology identified *Actinomyces* in 73% (11/15) of concretions and in 8/13 specimens in a separate series, while cultures frequently yield *Staphylococcus* (39%), *Streptococcus*, and *Nocardia*, and metagenomic sequencing shows anaerobe-dominated polymicrobial communities in all sampled concretions. (Evidence: [PMID: 41702562](#), [PMID: 22836798](#), [PMID: 28248874](#), [PMID: 34287258](#))

Finding 3 — Canaliculotomy with curettage is definitive; adjuncts improve outcomes

Canaliculotomy cured 98.34% of canaliculitis cases in a 302-case dataset; silicone tube intubation raised anatomical/functional success (100%/87.5% vs 78.3%/60.9%); long-term recurrence-free rate was 88% at 3.7 years. (Evidence: [PMID: 40067167](#), [PMID: 34275398](#), [PMID: 36431305](#))

Finding 4 — Broad etiologic spectrum; plugs are a key iatrogenic cause

Plug-related (secondary) canaliculitis constituted 18.3% of cases (all female, *Pseudomonas* dominant, 42%); fungal cases are led by *Candida* (27.1%) and *Aspergillus* (23.7%); viral cases are dominated by HSV (83.1%). (Evidence: [PMID: 34139956](#), [PMID: 42782278](#), [PMID: 41528827](#))

Finding 5 — Diagnosis is clinical (tetrad) plus UBM

The clinical tetrad and 80-MHz UBM (luminal ectasia + high-echo concretion) enable diagnosis; 85% of eyes had been previously misdiagnosed. (Evidence: [PMID: 36644466](#), [PMID: 31823059](#))

Finding 6 — Chronic, unilateral, lower-canalculus, adult disease

Lower canalculus 65%, upper 23%, both 12%; unilateral and chronic; pediatric onset extremely rare. (Evidence: [PMID: 22836798](#), [PMID: 31055896](#))

Finding 7 — Self-perpetuating stasis–colonization–concretion–inflammation cycle

Dry eye and odontogenic *Actinomyces* are key predisposing factors in immunocompetent hosts; involutional anatomical changes drive stasis; infection–inflammation forms a vicious circle. (Evidence: [PMID: 36927124](#), [PMID: 39488146](#), [PMID: 42366665](#))

Finding 8 — Conservative incision-sparing therapy is an effective first-line option

Incision-sparing management succeeded in 83.3% (10/12) with no recurrence; conservative expression + antibiotic irrigation is recommended initial therapy for *Actinomyces* disease. (Evidence: [PMID: 28576205](#), [PMID: 36927124](#), [PMID: 18580001](#))

Finding 9 — Rare disorder with excellent prognosis; watch for carcinoma masquerade

Actinomyces accounts for ~11% of primary cases; cure ≥88–98%; no disease-specific mortality; carcinoma can masquerade as canaliculitis, underscoring histopathology in atypical cases. (Evidence: [PMID: 36927124](#), [PMID: 29564416](#), [PMID: 16534063](#))

Mechanistic Model / Interpretation

The unifying model is a **biofilm/concretion-centered vicious cycle**. Chronic canaliculitis is best understood not as a simple bacterial infection but as a **niche disease**: age- and dry-eye-related tear stasis permits filamentous *Actinomyces* (often with anaerobic co-pathogens) to colonize and build a mineralized concretion that acts as a protected reservoir. This concretion explains the three most clinically important features of the disease: (1) its **chronicity and recurrence**, (2) its **resistance to topical antibiotics** (poor penetration of the biofilm/calculus), and (3) the **therapeutic imperative of physical removal**—cure tracks with completeness of concretion evacuation, not with antibiotic choice alone. Downstream, unresolved

inflammation produces canalicular dilatation and fibrosis/stenosis, which reinforce stasis and close the loop. Secondary (plug-related) disease short-circuits the upstream steps by supplying a ready-made foreign-body nidus.

Axis	Upstream → Downstream
Trigger	Dry eye / aging / plug → tear stasis
Microbial	Colonization → concretion/biofilm
Host response	Chronic epithelial inflammation → sequelae (dilatation, stenosis)
Clinical	Pouting punctum, discharge, epiphora
Therapy target	Remove concretion (curettage) ± antibiotics ± intubation

Evidence Base

PMID	Contribution	Type
22836798	Largest primary series: demographics, symptoms, localization, microbiology	Human clinical
41702562	Quantified <i>Actinomyces</i> as leading organism (73% of concretions)	Human clinical/histology
28248874	Concretion series confirming <i>Actinomyces</i> histopathology	Human clinical
34287258	Metagenomic sequencing of concretions; anaerobe predominance	Molecular/clinical
40067167	98.34% cure by canaliculotomy (large series)	Human clinical
34275398	Silicone intubation improves success	Human clinical (comparative)
36431305	88% recurrence-free long-term	Human clinical
34139956	Plug-related canaliculitis; <i>Pseudomonas</i>	Human clinical
42782278	Fungal lacrimal infection spectrum	Review
41528827	Viral (HSV-dominant) lacrimal infection spectrum	Review
36644466	Diagnostic clinical tetrad	Human clinical
31823059	UBM imaging signature	Human clinical/imaging
36927124	Dry eye + odontogenic <i>Actinomyces</i> ; conservative therapy	Human clinical
39488146	Infection–inflammation vicious-circle mechanism	Review
28576205	Incision-sparing management (83.3%)	Human clinical
16534063	Carcinoma masquerade — differential caution	Case report

Additional supporting reports on rare/novel organisms and pediatric disease: [PMID: 15764110](#), [PMID: 31246677](#), [PMID: 40234845](#), [PMID: 41429755](#), [PMID: 40788664](#), [PMID: 29426966](#), [PMID: 31055896](#), [PMID: 15167737](#).

Limitations and Knowledge Gaps

1. **Evidence quality:** All data derive from retrospective case series and case reports; there are few randomized trials and no population-based incidence/prevalence estimates. Selection and referral bias toward tertiary dacryology centers is likely.
 2. **No molecular/omics profiling:** No transcriptomic, proteomic, metabolomic, or lipidomic disease signatures exist; "omics" is limited to metagenomic pathogen identification.
 3. **No genetic architecture:** No causal genes, susceptibility loci, or modifier genes—appropriate for an acquired infection but limiting for knowledge-base fields designed for Mendelian disease.
 4. **No animal or in vitro disease models,** precluding mechanistic dissection of concretion formation and biofilm tolerance.
 5. **Under-diagnosis:** Chronic misdiagnosis means published cohorts underestimate true burden and skew toward severe/refractory presentations.
 6. **Overlap ambiguity:** The boundary between infectious chronic canaliculitis and "idiopathic canalicular inflammatory disease" ([PMID: 29373404](#)) is incompletely defined.
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Proposed Follow-up Experiments / Actions

1. **Prospective multicenter registry** with standardized microbiology (aerobic/anaerobic culture + 16S/metagenomics) to establish true incidence, organism frequencies, and recurrence rates.
2. **Randomized trial** of conservative incision-sparing management vs punctum-sparing canaliculotomy ± silicone intubation, powered for functional success and recurrence.
3. **In vitro biofilm/concretion model** co-culturing canalicular epithelial cells with *Actinomyces* to define mineralization triggers and test anti-biofilm agents.
4. **Concretion "omics":** shotgun metagenomics + metabolomics of concretions vs healthy lacrimal flora to characterize the pathogenic community and mineral matrix.
5. **Diagnostic pathway intervention study:** measure whether teaching the clinical tetrad + point-of-care UBM reduces diagnostic delay in primary/secondary eye care.
6. **Prospective evaluation of plug safety:** registry of dry-eye patients receiving punctal/canalicular plugs to quantify canaliculitis incidence and identify preventive protocols.

7. Histopathology mandate in atypical cases to systematically exclude canalicular carcinoma and other masquerades.

Report compiled from a 5-iteration autonomous investigation; 9 confirmed findings; 42 papers reviewed. Disease category: infectious disease. MONDO:0004924.

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